Input power connector
PWR_FLAG
VIN1
F1
0451005.MRL 5A
J1
ADAPTER 4.75–21V
PGND
Battery connector
J3
BATTERY 2S/4S
BATTERY
Output power connector
J2
VSYS OUTPUT
SYS
IC1
BQ25798RQMR
Q1
AON7528
Q2
AON7528
D1
12V
R1
270R
NT4
VBUS
boost side
buck side
L1
2.2uH
J4
MCU / BIAS INTERFACE
~INT
CHG_REQ
I2C_SCL
I2C_SDA
MAIN_3V3
REGN
R8
2k2
LED
D2
C3
4.7u 25V
NT2
ILIM_HIZ
R6
5k6
R7
33k
NTC temp sensor
J6
Grounds
R12
2R 5% pulse
C26
2.2u 100V X7R
USB-C PROGRAM / NATIVE USB
U1
ESP32-C3-MINI-1-N4X
TPD2EUSB30A (PROVISIONAL)
SW1
BOOT / DOWNLOAD
SW2
RESET
NORMAL 3V3 / SERVICE USB 3V3
U3
TPS62933DRLR NORMAL 3V3
U4
AP7361C-33E-13 USB 3V3 LDO
POWER MUX – MAIN PRIORITY
U5
TPS2116DRL NORMAL PRIORITY
I2C / INT DOMAIN ISOLATION
U6
TCA9517DGKR REV.0
Q8
PMV16UN MAIN_VALID
Q9
PMV16UN TCA_EN
CHARGING: MAIN PRESENT AND FIRMWARE REQUEST
Q4
BC847 INT DOMAIN ISOLATOR
Q6
BC847 MAIN_VALID SINK
Q5
BC857 MAIN-GATED CHG SOURCE
Q7
BC847 MCU CHG ARM SINK
H1
H2
H3
H4
U7
TPS62933DRLR
U8
LM74700QDBVRQ1
J8
5V / GND
F5_5V_OUT

F1 prototype: Littelfuse 0451005.MRL, 5 A, 125 Vdc; qualify temperature, cable, hot-plug, polarity-FET SOA and coordination.

VIN damping R12 2 ohm / C26 2.2uF 100V: selected prototype parts; VAC1 hot-plug qualification OPEN.

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

PGND = system return; GND = quiet TS / PROG / NTC; NT1 local tie; no PCB moat.

VIN hot-plug RC: TI Fig. 8-1 (>15V)
Provisional R12 / C26; validate on prototype

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

SERVICE USB / ESP32-C3 RECOVERY – prototype, not fabrication approval

U3 SYS -> MAIN_3V3 buck; U4 USB_VBUS -> USB3V3 LDO; U5 normal-priority mux -> MCU3V3.

USB-only ROM flashing: BOOT held low during RESET; no battery/adapter required. Wi-Fi off in service firmware.

Complete power interruption: EN 10k/1u RC and GPIO straps restart MCU. Transition resets acceptable; hardware tests pending.

U1 project footprint corrected to 61 lands / 53 numbers; verify RF keepout, stencil and placement in PCB phase.

Q5/Q6/Q7 default off; Q3 sinks CE only on request. Q6 is not a precision MAIN voltage monitor.

CE pull-up via REGN/MAIN diode OR. BQ25792 2S charger-only reset/default 2A risk OPEN.

PRIMARY SERVICE: native USB-C Serial/JTAG for programming, logging and debug; USB-only MCU power.

OPTIONAL EMERGENCY UART: TP5 ESP_TX GPIO21, TP6 ESP_RX GPIO20, TP7 GND; no fitted header.

PCB edge, 2.54mm pitch TOP view; triangle at TP5. UART high-Z while target off; NO power input.

SEPARATE MEASUREMENT ONLY: TP8 MCU3V3, TP13 MAIN_3V3; NOT in UART group. BOOT/RESET buttons retained.

TP14 CE_N: separate passive charger-domain bring-up probe. No added RESET/BOOT/I2C pads.

POWER MEASUREMENTS: TP15 VIN1, TP16 BATTERY, existing TP4 SYS; separate from MCU pogo group.

TP17 F5 OUTPUT: measurement only. No testpad is an authorized power input.

F5 OPTIONAL 5V / 2A: populate ALL U7 U8 Q10 L3 R60–R63 C60–C69 J8; F0 omit all.

SYS 6–18V -> 5.04V buck -> ideal diode -> J8 pin1 +5V, pin2 PGND. No MCU/USB control.

Sheet: /
File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

Size: A1	Date:	**Rev: 2.1**
KiCad E.D.A. 10.0.7		Id: 1/1